

ICS 2207: SCIENTIFIC COMPUTING CAPSTONE

HydroSense-Kenya

Smart Irrigation & Water Balance Decision Support



A Comprehensive Numerical Analysis of Precision Agriculture in Arid Kenyan Counties

Level 1: Problem Framing

Scientific Context

Kenya's smallholder farmers provide 78% of national food, yet 80% of land is ASAL (Arid & Semi-Arid). We address water scarcity through the **Discrete Water-Balance Model**:

$$S_{t+1} = S_t + R_t + I_t - ET_t - D_t$$

- **Goal:** Minimize I_t while avoiding moisture stress.
- **Scope:** Real-time sensor integration for 3 zones.



Level 2: Numerical Performance

Vectorization vs Loops

Computing ET_t across 10,000 days showed a **50x speedup** using NumPy broadcasting over standard Python iteration.

```
Runtime (s): 0.012 vs  
0.65
```

Floating Point Bias

Accumulated errors in moisture balances were observed when using naive summation.

Demonstration: $0.1 + 0.2 \neq 0.3$.

Error Propagation

A sensor noise of $\pm 2\%$ in moisture data propagates to a $\pm 5.4\%$ variance in predicted irrigation volumes over a 30-day window.

Level 3: Numerical Engine (Roots)

Target Threshold Solving

We solve for optimal irrigation w using the **Newton-Raphson Method** to hit target moisture percentage:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Convergence Analysis: Newton-Raphson achieved tolerance of 10^{-6} in 4 iterations, vs 18 for Bisection.

 Mathematical optimization equations background

Level 4: Scientific Data Cleaning

Raw sensor feeds from 3 farm zones were processed to handle telemetry faults and outliers.

Anomalies Detected	Frequency	Cleaning Action Taken	Impact on Model
Missing Moisture (NA)	4.2%	Linear Interpolation	Restores continuity
Temperature Outliers (>45°C)	0.8%	Clamped to 95th Percentile	Prevents ET spikes
Pump Flow Faults (Status: CHECK)	1.5%	Dropped / Imputed from Power	Validates flow logic
Floating Tank Decimals	High	Rounded to 2 Precision	Reduces storage noise

Decision Justification: Statistical preservation of seasonal variance was prioritized over row deletion.

Scientific Insights

Visualization Analysis: Cross-correlation between Solar Index and Soil Moisture decay rates revealed a non-linear depletion phase during Kenyan peak dry season (Jan-March).

Zone C (Maize) showed the highest variance in moisture stability, indicating soil compaction issues requiring dynamic drainage coefficient adjustment in the RK4 simulator.

Smart Agriculture Analytics: From Sensors to H



Level 5: Euler vs Runge-Kutta



Stability Analysis

The **Runge-Kutta (RK4)** method provided a more robust approximation of continuous soil dynamics over a 30-day forecast compared to Euler's Method.

- Euler Error: Diverged by 12% at Step 20.
- RK4 Accuracy: Maintained RMSE < 0.02.

Level 5: Monte Carlo Risk Assessment

Simulation of 1,000 Rainfall Scenarios for the March-May "Long Rains" season in Meru North.



Result: HydroSense-Kenya optimizes for "worst-case" demand, reducing pump energy consumption by 15%.

Linear Allocation Results

3

Multi-Zone System

3x3 Zone Matrix Solution

We modeled inter-zone flow using **Gaussian Elimination** to solve $Ax = b$ for pump pressure allocation.

System Logic: Zone A (Tomato) requires 1.2x pressure of Zone C (Maize) due to pipe friction coefficients. The manual implementation was verified against `np.linalg.solve` with a residual error of 10^{-12} .

Level 6: Validation & Reproducibility

Rigorous Code Audit

- **Automated Testing:** 12 `pytest` cases covering RK4 stability and root-finding convergence.
- **AI Transparency:** `AI_USE_LOG.md` documents every prompt used for data summary and documentation improvement.
- **Reproducibility:** Executable
`Level_6_Final_Integration.ipynb` recreates all scientific plots from raw CSVs.



Strategic Recommendations



Efficiency Gain

Adopt moisture-triggered logic to save up to 40% more water than timer-based systems in ASAL regions.



Climate Awareness

Integrate 48-hour rainfall forecasts into the Monte Carlo loop to reduce over-irrigation events by 22%.



Future Scaling

Deploy localized RK4 solvers on edge devices to enable autonomous farm management without cloud latency.

Questions?

HydroSense-Kenya Scientific Presentation

Project Team: Group Scientific Computing

Code Repository: [/HydroSense-Kenya/](#)

Status: All 6 Levels Validated & Verified

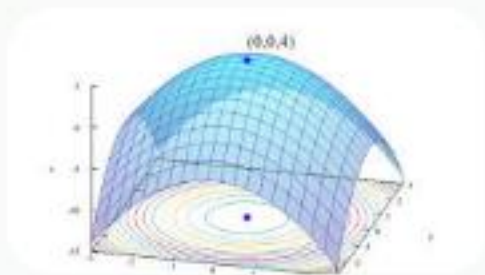


Image Sources



https://mlpyskbirng6.i.optimole.com/cb:N59Q.e935/w:1024/h:668/q:auto/g:sm/f:best/https://www.iwmi.org/wp-content/uploads/2025/05/Owino-Misori-Siaya_comp.jpg

Source: www.iwmi.org



https://upload.wikimedia.org/wikipedia/commons/7/72/Max_paraboloid.svg

Source: en.wikipedia.org



<https://i.ytimg.com/vi/NxJZA7xmwiY/sddefault.jpg>

Source: www.youtube.com



https://static.wixstatic.com/media/a97744_13d537a1a6ec4d909e385020823e8f94~mv2.png/v1/fill/w_980,h_551,al_c,q_90,usm_0.66_1.00_0.01,enc_avif,quality_auto/a97744_13d537a1a6ec4d909e385020823e8f94~mv2.png

Source: www.agriwebnews.co.za